

Key Considerations When Selecting A Robotic Soldering System

CRITERION CATEGORY	KEY FACTORS TO EVALUATE	WHY IT MATTERS FOR ELECTRONICS MANUFACTURING
Technical Performance	Positioning accuracy (e.g., $\pm 0.02\text{-}0.05\text{mm}$)	Critical for fine-pitch components and high-density boards
	Repeatability	Ensures consistent joint quality across runs
	Axis configuration (3-6 axes)	More axes help access complex PCB geometries
	Temperature control range and precision	Maintains correct heat profiles for different alloys
Vision and Sensing	Vision-guided alignment (2D/3D)	Reduces setup errors and compensates for PCB placement variances
	Tip pressure and position feedback	Prevents board damage and improves solder quality
Production Fit	Board size/work envelope	Must cover current and future PCB formats
	Throughput capability (joints/hr)	Aligns robot speed with production takt
	Flexibility for mixed production	Easier changeover between SKUs
Software and Integration	Ease of programming and UI	Faster onboarding, reduces reliance on specialists
	MES/PLC connectivity	Enables traceability and smart factory integration
Standards and Compliance	IPC-J-STD-001/IPC-A-610 adherence	Ensures reliable solder joints meeting industry quality norms
	CE/UL/ESD safety compliance	Required for global electronics manufacturing environments
Safety and Environment	Integrated fume extraction	Protects workers and meets workplace health standards
	ESD safety features	Critical for sensitive electronic components
Cost and Lifecycle	Total cost of ownership (TCO)	Includes maintenance, consumables, energy
	Warranty and service support	Reduces downtime risk and supports long-term use
Quality and Traceability	Real-time joint monitoring and logging	Supports SPC and reduces rework
	Defect detection and reporting	Improves yield and helps continuous improvement

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APOLLO SEIKO J-CAT Wave

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THANK YOU TO OUR ESTEEMED ADVISORY PANEL

We extend our sincere gratitude to our esteemed Advisory Panel—

Mr B.N. Shukla (Jabil Circuit India Pvt Ltd), **Mr Shrikant Borkar** (Intelux Electronics Pvt Ltd), **Mr Mukesh Vasani** (Aimtron Electronics Limited), **Mr Ashvin Navadia** (Vinrox Technologies), and **Mr Narayan Kumar J.** (Sanjay Technologies Pvt Ltd)—for graciously being part of our Advisory Panel and for their constant guidance and support.